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Multimodal graph neural networks for earth observation and sustainable resource management: a comprehensive review and research roadmap

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Abstract
Multimodal Graph Neural Networks (GNNs) have emerged as a powerful paradigm for integrating heterogeneous Earth Observation (EO) data-such as optical and SAR imagery, in-situ sensor readings, geospatial vector layers, and socio-economic records-into unified relational representations. By jointly modeling complementary spatial and temporal modalities, multimodal GNNs enable more accurate, robust, and interpretable solutions for environmental monitoring and sustainable resource management. This article presents, to the best of the authors' knowledge, the first comprehensive survey dedicated to multimodal GNN methods for EO applications, covering literature up to 2025. A detailed taxonomy of fusion strategies is proposed, including early, late, hybrid, cross-modal attention, and foundation-model-based approaches, and representative architectures are analyzed across agriculture, water resources, forestry, and urban infrastructure. Publicly available datasets, benchmark tasks, and evaluation metrics are systematically compared, with critical issues such as label scarcity, domain shift, scalability, and real-time deployment examined in depth. Special emphasis is placed on explainability, uncertainty quantification, and energy-efficient "green AI" practices. Finally, open research challenges are identified and a forward-looking roadmap is outlined toward standardized benchmarks, reproducible experimental protocols, and operational multimodal GNN systems capable of delivering actionable, policy-relevant EO insights at scale.

Keywords Multimodal graph neural networks, Earth observation, Data fusion, Sustainable resource management, Remote sensing, Spatio-temporal modeling

1 Introduction

Sustainable management of environmental and socio-economic systems has become an urgent global priority in the face of climate change, population growth, rapid urbanization, and overexploitation of natural resources. Meeting these challenges requires robust monitoring and decision-support systems capable of capturing the complex interactions among environmental, infrastructural, and human systems. Earth Observation (EO)

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technologies-encompassing satellite remote sensing, aerial platforms, and in-situ sensor networks-now generate unprecedented volumes of data across diverse spatial, temporal, and thematic scales. These heterogeneous data streams provide complementary perspectives: optical and synthetic aperture radar (SAR) imagery capture large-scale surface and structural patterns, while ground-based sensors deliver precise point measurements of environmental variables such as soil moisture, precipitation, or pollutant concentrations [1, 2]. Geospatial vector datasets, including road networks, administrative boundaries, and land-use maps, contribute explicit spatial topology, and textual or tabular datasets convey socio-economic, policy, or event-related information [3, 4]. Taken together, these sources form a rich but fragmented view of the Earth system.

Global sustainability goals cannot be addressed effectively without harnessing the full multimodality of EO data. The integration of these heterogeneous modalities-commonly referred to as multimodal fusion-is increasingly recognized as a key enabler for advancing EO-based decision-making. Fusion across modalities enhances robustness to missing or noisy data, increases predictive accuracy, and enables richer, more holistic representations of complex socio-environmental systems [1, 4, 5]. For example, in agriculture, combining multispectral satellite imagery with meteorological sensor readings and irrigation infrastructure maps improves yield prediction and stress detection [6, 7]. In hydrology, integrating radar remote sensing with in-situ gauge networks and digital elevation models has strengthened flood forecasting and groundwater modelling [2, 8]. In urban contexts, fusing high-resolution imagery, IoT-based energy consumption data, and vector-based transport networks facilitates monitoring of emissions, optimization of infrastructure use, and identification of vulnerable areas [9, 10]. These examples illustrate the central role of multimodality in transforming raw EO observations into actionable insights for sustainability. Yet multimodal fusion is technically challenging. Data arrive at different spatial and temporal resolutions, with varying coverage, quality, and reliability. Optical imagery is often degraded by cloud cover, SAR can suffer from speckle noise, and IoT networks may experience outages or incomplete coverage [1, 11]. Aligning these heterogeneous sources-such as linking pixel-based imagery with node-based sensor networks or text-based event reports-requires models capable of representing both spatial structure and cross-modal relationships [12, 11].Traditional machine learning methods, such as convolutional neural networks (CNNs) and transformers, have achieved great success in unimodal tasks like image recognition or sequence modelling. Early contributions in multimodal EO and AI integration provide the historical context for these developments. For instance, [13] introduced M^3Fusion, one of the first deep learning architectures for multiscale multimodal temporal satellite data fusion, while [14] highlighted how deep learning and EO jointly support the Sustainable Development Goals. Conceptual frameworks such as [15] integrated artificial intelligence into the EO4GEO body of knowledge, shaping interdisciplinary curricula. Beyond EO, advances in multimodal learning in robotics [16] and neuroscience modeling [17] demonstrated the feasibility of combining heterogeneous data sources, providing methodological foundations later adapted to Earth Observation. Nevertheless, despite their impact, CNNs and transformers remain limited when it comes to encoding irregular spatial topologies, relational dependencies, and cross-modal interactions that characterize EO data. CNNs operate on regular grids, making it difficult to incorporate non-Euclidean structures

such as road networks or river basins, while transformers often require massive training data and may ignore geospatial topology unless explicitly engineered.

Graph Neural Networks (GNNs) have emerged as a promising paradigm to bridge this gap. By representing entities as nodes and their relationships as edges, GNNs naturally encode the topology of geospatial networks, sensor graphs, and object-to-object interactions [18, 11]. Message-passing and attention mechanisms enable the integration of information across neighborhoods, thereby capturing both local and global dependencies [12, 19]. This makes GNNs well suited for EO applications such as road network analysis, river basin modelling, deforestation monitoring, and crop yield forecasting [6, 8, 20]. When extended with multimodal fusion strategies, GNNs can integrate heterogeneous node and edge features from diverse sources, producing models that jointly exploit spectral, spatial, temporal, and contextual information [5, 21]. Fusion approaches vary in design: early fusion combines raw features before graph construction, late fusion integrates outputs of modality-specific encoders, and hybrid methods employ cross-modal attention or heterogeneous graph structures to dynamically propagate and weight multimodal signals [1, 4, 11]. More recently, foundation-model-based strategies have emerged, where pretrained vision, text, or geospatial models are incorporated into GNN pipelines to improve generalization in data-scarce settings [22, 7]. These advances highlight why GNNs, compared to other learning architectures, are particularly well aligned with the relational, heterogeneous, and multimodal nature of EO data. The main contributions of this survey are summarized as follows:

- Provide the first comprehensive review of multimodal Graph Neural Networks (GNNs) for Earth Observation (EO), emphasizing sustainability-oriented applications and studies up to 2025.
- Present a structured taxonomy of multimodal fusion strategies-early, late, hybrid, cross-modal attention, and foundation-model-based designs.
- Examine representative architectures across agriculture, hydrology, forestry, and urban systems, highlighting domain-specific adaptations, strengths, and limitations.
- Summarize publicly available datasets, benchmark tasks, and evaluation metrics, identifying major technical challenges including label scarcity, domain shift, scalability, interpretability, and energy efficiency.
- Outline a forward-looking research roadmap covering self-supervised learning, adaptive graph structures, edge/cloud deployment, and the need for standardized, reproducible benchmarks.

The remainder of this paper is organized as follows. Section 2 introduces the foundations of multimodal GNNs in Earth Observation. Section 3 reviews applications across key EO domains. Section 4 presents benchmark datasets and preprocessing pipelines. Section 5 discusses multimodal fusion strategies in GNN architectures. Section 6 outlines current challenges and open research problems, while Sect. 7 highlights forward-looking research directions. Finally, Sect. 8 concludes with implications for operational, policy-relevant applications of multimodal GNNs in sustainability.

2 Foundations of multimodal graph neural networks in earth observation

Graph Neural Networks (GNNs) offer a principled framework for learning on relational structures by representing entities as nodes and their interactions as edges. This capability is particularly relevant in Earth Observation (EO), where environmental processes and human activities exhibit inherently networked and spatial–temporal dependencies-examples include river basins, road networks, ecological corridors, and sensor deployments. Several recent perspectives have underscored the value of graph-based representations for moving beyond grid-centric processing of EO data, enabling models that capture irregular spatial relations and multi-sensor dependencies [18, 12, 11]. EO data are intrinsically multimodal. Common sources include optical multispectral imagery, synthetic aperture radar (SAR), LiDAR point clouds, meteorological and hydrological time series, IoT-based in-situ measurements, geospatial vector datasets (e.g., roads, cadastral parcels), and textual or tabular records (e.g., reports, census statistics). Each modality differs in spatial and temporal resolution, acquisition frequency, noise profile, and semantic content. Combining such complementary information improves robustness, reduces ambiguity, and enables richer sustainability-focused task formulations [1, 4, 23]. Dedicated multimodal EO datasets have recently emerged to facilitate this integration, reflecting a broader shift toward multi-sensor evaluation resources [24, 23].

Designing a multimodal GNN for EO typically requires addressing three interlinked questions:

1. *Graph construction* How to define nodes and edges from heterogeneous inputs.
2. *Feature encoding and fusion* How to extract modality-specific representations and combine them effectively.
3. *Message passing and temporal modeling* How to propagate and aggregate information across space and time.

Nodes may correspond to pixels or patches, object segments, administrative units, or physical sensors; edges can encode spatial adjacency, hydrological flow, functional similarity, or learned affinities. Heterogeneous graph structures that preserve distinct node and edge types are advantageous when modalities differ fundamentally in semantics [11, 5].

The generic pipeline for multimodal graph neural networks (GNNs) in Earth Observation (EO) is illustrated in Fig. 1. This pipeline demonstrates the integration of various modalities to enhance feature representation and improve model performance. Table 1 outlines representative GNN architectures and their suitability for common EO tasks.

2.1 Fusion strategies

Multimodal fusion within GNN pipelines can be categorized by the stage at which modalities are combined. In EO, common strategies include:

- *Early fusion* Concatenation of raw or low-level features before graph assembly.
- *Late fusion* Independent modality encoders with fusion of embeddings or predictions.
- *Hybrid fusion* Interleaved fusion within intermediate network layers.
- *Cross-modal attention* Learnable weighting of modality contributions at node or edge level.

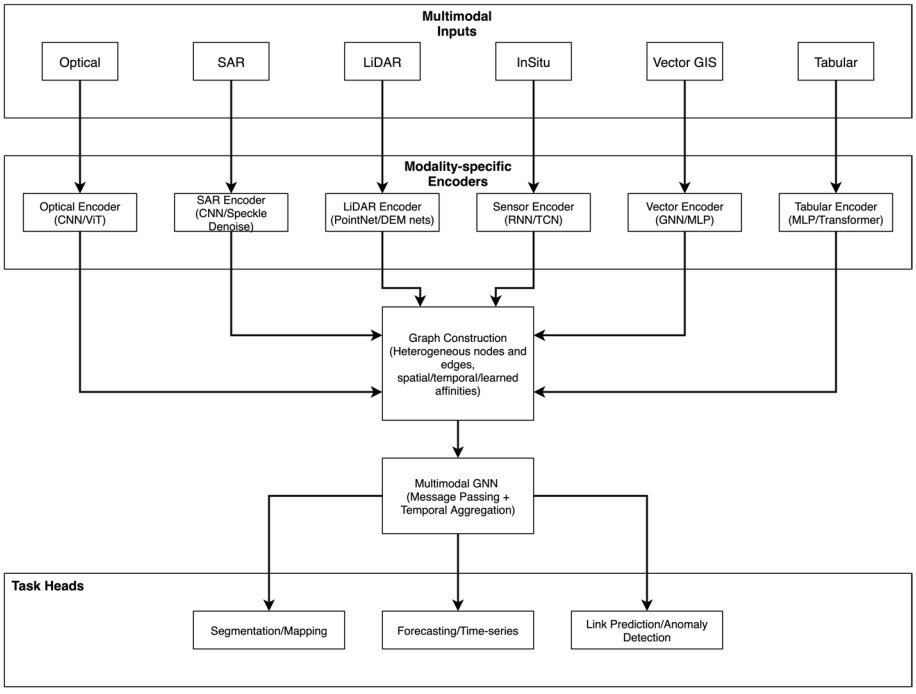

Fig. 1 Generic pipeline for multimodal GNNs in EO

Table 1 Representative GNN architectures and their applicability to EO tasks

Architecture	Core mechanism	Typical EO applications / notes
Graph Convolutional Network (GCN)	Neighborhood averaging / spectral filters	Land-cover smoothing, region-level classification; simple and scalable [18].
Graph Attention Network (GAT)	Attention-weighted aggregation	Identifies important neighbours/modalities; suited for heterogeneous sensor networks [11].
Spatio-Temporal GNN (ST-GNN)	Temporal modules with message passing	Flood forecasting, groundwater dynamics, crop growth modeling [6, 8].
Heterogeneous GNN (R-GCN / HAN-like)	Type- and relation-aware propagation	Preserves node types for integrating sensors, parcels, and vector networks [11, 5].
Hierarchical / Patch-Graph GNN	Multi-scale pooling	Merges patch-level imagery with regional graphs for change detection [23].
Inductive / Sampling GNN (GraphSAGE)	Neighbourhood sampling for scalability	Enables large-area monitoring and streaming data updates [18, 25].

- *Foundation-model integration* Incorporation of large pretrained encoders into GNN pipelines.

Choice of strategy is influenced by data alignment quality, availability of labelled samples, and computational constraints [4, 7, 5]. Table 2 compares these approaches.

2.2 Spatio-temporal alignment and modalities

A key practical challenge is aligning modalities with differing acquisition schedules and spatial footprints. Optical imagery may suffer from cloud cover, SAR acquisitions follow fixed revisit cycles, and in-situ sensors often record at irregular intervals. Graph formulations can mitigate these mismatches using affinity-based edges, temporally aware connections, or multi-rate temporal modules [1, 8]. Table 3 summarizes typical EO modalities, their resolutions, and fusion considerations.

Table 2 Fusion strategies - strengths, weaknesses, and EO suitability (illustrative)

Fusion type	How it works	Advantages	Limitations
Early fusion	Concatenate raw / low-level features	Simple; joint feature learning	Sensitive to misalignment, scale mismatch
Late fusion	Fuse embeddings or outputs	Robust to missing modalities; modular	May lose fine-grained cross-modal interactions
Hybrid fusion	Interleave encoders and message passing	Balances global and local interactions	Increased complexity and tuning cost
Cross-modal attention	Learn importance weights per modality/node	Adaptive to variable signal strength	Requires more data; higher computational load
Graph-of-patches	Graph from image patches + sensors as nodes	Preserves spatial context with sensor linkage	Larger graphs; pooling required
Foundation-finetune	Pretrained encoders integrated into GNN	Better transfer and generalization	Pretraining cost; possible domain mismatch

Table 3 Common EO modalities, typical resolutions, and fusion considerations (illustrative)

Modality	Typical resolution	What it adds	Key fusion note
Optical multispectral	3–30 m (Planet, Sentinel-2, Landsat)	Vegetation/land-cover indices	Gaps from clouds; pair with SAR or interpolation [23]
SAR	1–20 m (Sentinel-1, etc.)	All-weather backscatter, structure	Speckle noise; complements optical [24]
LiDAR / DEM	Sub-meter–tens of meters	Elevation, canopy height, fine structure	Sparse; useful as structural prior [4]
In-situ sensors / IoT	Point-based, variable cadence	Ground-truth time series	Link via proximity or flow; handle missing data [2]
Vector layers	Arbitrary geometry	Explicit topology (roads, parcels)	Direct mapping to graph nodes/edges [20]
Tabular / textual	Region-level / unstructured	Socio-economic, policy context	Needs embedding or KG integration [22, 10]

2.3 Why GNNs for multimodal EO?

While convolutional and transformer-based architectures have achieved significant success in unimodal Earth Observation (EO) tasks, several intrinsic characteristics of multimodal EO data make Graph Neural Networks (GNNs) particularly advantageous. EO datasets frequently encode irregular and relational structures-such as transport networks, hydrological basins, ecological corridors, or administrative boundaries-that grid-based convolutional operations cannot capture effectively. Moreover, multimodal fusion requires the joint processing of heterogeneous and asynchronous sources including optical imagery, synthetic aperture radar (SAR), LiDAR, socio-economic statistics, and IoT sensor data. In such contexts, GNNs provide a principled framework for message passing across nodes and edges, enabling relational reasoning across modalities, scales, and temporal dimensions [26, 27, 14]. Unlike CNNs, which rely on fixed spatial grids, or transformers, which depend on dense attention across tokens, GNNs can explicitly represent spatial topology and inter-object dependencies, preserving both local and global context. This flexibility allows them to model complex environmental systems where interactions are non-Euclidean and data availability is uneven. In practice, GNNs have demonstrated measurable benefits across multiple EO domains. In *agriculture*, multimodal GNNs integrate multispectral imagery with parcel boundaries, soil and weather data, providing improved crop-yield forecasts and stress detection [6]. In *hydrology*, spatio-temporal GNNs combine SAR imagery, digital elevation models, and river-gauge records to enhance flood prediction and groundwater modeling [8]. In *forestry*, they fuse optical imagery, radar backscatter, and ecological vector layers for deforestation

monitoring and habitat mapping [28]. In *urban systems*, heterogeneous GNNs merge high-resolution imagery, census data, and mobility networks to classify land use, estimate energy demand, and assess infrastructure vulnerability [10]. These examples demonstrate that GNNs serve as a bridge between the relational structure of Earth systems and the multimodal data that describe them. By unifying spatial, temporal, and contextual information, GNN-based fusion not only improves predictive performance but also enhances interpretability and robustness-making GNNs a natural foundation for the next generation of multimodal EO analytics.

2.4 Interpretability, uncertainty, and scalability

For sustainability-related EO applications, interpretability and uncertainty quantification are critical. Recent work has explored uncertainty-aware fusion and interpretable mapping to improve trust in model outputs [10, 28]. Responsible AI guidelines for EO highlight transparency, reproducibility, and lifecycle considerations, including computational efficiency and environmental footprint [29]. Scaling multimodal GNNs to continental or global applications remains challenging, motivating the use of sampling-based GNNs, hierarchical pooling, and inductive training approaches [25, 23]. Despite progress, benchmarking remains fragmented. While several multimodal EO datasets and tasks exist, a standardized suite for multimodal GNN evaluation is still lacking [24, 23]. GNNs thus provide a flexible and unifying foundation for multimodal EO fusion, enabling explicit modelling of spatial, temporal, and cross-modal relationships. The next section introduces a structured taxonomy of fusion strategies, followed by domain-specific applications and benchmarking considerations.

3 Applications of multimodal graph neural networks in earth observation

The use of Multimodal Graph Neural Networks (GNNs) in Earth Observation (EO) has expanded rapidly across environmental, agricultural, hydrological, forestry, and urban domains. By fusing complementary data sources within a graph-based framework, these models exploit both the relational structure of environmental systems and the unique strengths of each sensing modality. Compared to grid-based approaches, graph formulations enable explicit representation of spatial topology, cross-modal dependencies, and temporal evolution [18, 12, 11].

Land-cover and land-use mapping has benefited from integrating spectral, structural, and contextual information. Very-high-resolution optical imagery, LiDAR-derived canopy and elevation metrics, and cadastral vector parcels are fused within patch- or region-level graph structures to produce spatially coherent classifications and reduce salt-and-pepper noise [30, 10, 20].

Change detection for urban expansion, deforestation, or coastal transformation often combines multi-temporal optical and SAR imagery with DEMs. Graph- or patch-graph approaches encode spatio-temporal neighbourhoods, enabling robust detection of subtle or gradual changes despite seasonal and sensor variability [23, 20, 31].

Agricultural monitoring leverages multispectral time series, meteorological station records, and parcel connectivity graphs. Spatio-temporal GNNs and transformer-based encoders have been applied to forecast yields, detect crop stress, and generate site-specific management recommendations [6, 32].

Hydrology and water resource management applications integrate radar/SAR imagery, river-gauge time series, and terrain models for flood forecasting, groundwater modelling, and water quality assessment. Temporal message passing in ST-GNN architectures captures hydrological flow dynamics and seasonal patterns [2, 8, 23].

Forestry, biodiversity, and hazard assessment employ multimodal GNNs to combine optical and SAR imagery with ecological and topographic vectors for mapping deforestation, habitat fragmentation, and landslide susceptibility. Some approaches incorporate domain priors through graph contrastive learning or knowledge-guided models [1, 33, 28].

Urban sustainability and infrastructure studies integrate high-resolution imagery, mobility and transport data, and census or smart-meter readings using heterogeneous GNNs. Applications include energy-demand estimation, traffic–emissions coupling, and functional-region classification [9, 25, 34]. Recent graph-based advances have further extended multimodal GNNs toward building extraction and domain adaptation. For example, Li et al. [35] introduced a joint semantic–geometric learning framework for polygonal building segmentation from high-resolution imagery, while Zheng et al. [36] proposed an evidential graph contrastive alignment method for source-free domain adaptation, both contributing to improved spatial reasoning and transferability in EO applications.

Several published case studies further demonstrate the measurable benefits of multimodal GNNs in EO. In agriculture, Sarkar et al. [6] applied a multimodal meta-transformer with temporal GNNs for crop yield prediction, reporting an R^2 of 0.84 and a 12% reduction in RMSE compared to unimodal baselines. In hydrology, Taccari et al. [8] combined SAR imagery, DEMs, and in-situ gauges within a spatio-temporal GNN for groundwater modeling, achieving a 15% improvement in Nash–Sutcliffe Efficiency over CNN-based methods. For urban land-use mapping, Yan et al. [10] integrated optical imagery with socio-economic and census data via an uncertainty-aware GNN, obtaining an overall accuracy of 91.3% and an F1-score of 0.89. Similarly, Liu et al. [28] showed that knowledge-guided GNNs improved deforestation mapping in Southeast Asia, achieving an IoU gain of 8% over CNN-based baselines. These case studies highlight that multimodal GNNs not only capture complex spatial–temporal dependencies but also deliver tangible improvements in predictive accuracy, robustness, and interpretability across real-world EO applications. A broader overview of representative applications and typical modality combinations is provided in Table 4.

Table 4 Representative applications of multimodal GNNs in EO

Domain	Typical modalities	GNN Role / Approach	Ref
Land cover / land use	Optical multispectral, LiDAR/DEM, cadastral vectors	Node/region classification; hierarchical patch-graph GNNs	[30, 10, 20]
Change detection	Multi-temporal optical + SAR, DEM	Temporal embeddings, graph matching, patch-graph pooling	[23, 20, 31]
Agriculture	Multispectral time series, weather stations, parcel graphs	Spatio-temporal GNNs, temporal attention, meta-transformers	[6, 32]
Hydrology	Radar/SAR, river gauges, DEM, land cover maps	ST-GNNs for forecasting and anomaly detection	[2, 8]
Urban sustainability	High-res imagery, mobility data, census/energy meters, OSM	Heterogeneous GNNs, relational embeddings	[9, 25, 34]
Urban infrastructure & heritage	Optical imagery, SAR (MT-InSAR), socio-economic / historical records	Infrastructure risk prediction; heritage preservation with EO and deep learning	[37, 38]

Across domains, recurring challenges include limited labelled data, domain shift between regions and seasons, and the operational need for calibrated uncertainty estimates. Emerging work in explainability and responsible AI for EO emphasises transparency, reproducibility, and lifecycle impacts alongside technical performance [39, 10, 29].

3.1 Practical EO systems using multimodal GNNs

While much of the current literature focuses on experimental architectures and benchmark datasets, recent work has begun to demonstrate the operational potential of multimodal Graph Neural Networks (GNNs) within real-world Earth Observation (EO) systems. These systems integrate heterogeneous data streams-satellite imagery, sensor networks, and socio-economic records-to support data-driven decision-making in key sustainability domains.

In *agriculture*, multimodal GNNs have been deployed in precision-farming platforms by combining multispectral satellite imagery, parcel boundaries, and meteorological data to deliver field-level yield forecasts and early crop-stress detection. Such systems enable more efficient irrigation, fertilizer management, and food security planning [6].

In *urban planning and infrastructure management*, heterogeneous GNNs are being used in EO-based platforms that fuse high-resolution imagery, mobility data, and census information to support land-use classification, emission monitoring, and infrastructure risk assessment. These approaches allow urban authorities to map vulnerable regions, optimize energy use, and guide sustainable city planning [10].

In *water resource monitoring*, spatio-temporal GNNs integrate radar and SAR imagery with hydrological gauges and digital elevation models, enabling near real-time flood forecasting and groundwater variability assessment. These systems are increasingly being incorporated into disaster preparedness and resource allocation frameworks [8].

Together, these examples illustrate how multimodal GNNs are evolving from academic research prototypes to practical components of operational EO systems. By explicitly modeling spatial topology and cross-modal dependencies, they provide interpretable and actionable insights that directly support agriculture, urban sustainability, and water resource resilience.

3.2 Data usage, tasks, and multimodality in EO-GNN research

In the context of this survey, multimodality in Earth Observation (EO) is defined in a broad sense, extending beyond conventional remote sensing imagery to include diverse and complementary data sources. While satellite-based modalities such as multispectral and hyperspectral optical imagery, synthetic aperture radar (SAR), and LiDAR-derived digital elevation models remain central, many recent studies have demonstrated the added value of integrating ancillary data such as OpenStreetMap (OSM) layers, cadastral or administrative vector datasets, census and socio-economic indicators, meteorological and hydrological time series, and IoT-based in-situ sensor measurements. These heterogeneous inputs differ in spatial and temporal resolution, acquisition frequency, and semantic content, yet together they provide a more holistic representation of environmental systems and human activity. Within such multimodal settings, graph neural networks (GNNs) have been applied to a wide range of EO tasks. Classification problems, for instance, include land-cover and land-use mapping, crop-type identification, and deforestation monitoring, where the relational structure of parcels or patches helps

overcome noise and heterogeneity in raw imagery. Segmentation and detection tasks, such as change detection in urban expansion, deforestation alerts, or coastal transformation, benefit from multimodal fusion by combining optical, SAR, and elevation data to highlight subtle transitions that may not be apparent in a single modality. Forecasting and prediction tasks represent another active area, where temporal GNN models have been used for crop yield forecasting, hydrological flow prediction, and flood or drought forecasting by linking optical time series with meteorological and sensor data. Finally, anomaly detection and vulnerability assessment tasks increasingly incorporate multimodal graphs for applications such as landslide susceptibility mapping, infrastructure risk monitoring, and urban vulnerability assessment, where contextual information from socio-economic or structural datasets augments physical observations.

Table 5 summarizes common tasks, the modalities involved, and representative works.

4 Benchmark datasets and preprocessing pipelines

Benchmark datasets are essential for developing, evaluating, and comparing multimodal Graph Neural Networks (GNNs) and other AI models across domains such as Earth observation, environmental monitoring, urban analytics, and resource management. The choice of dataset and the design of preprocessing pipelines directly affect model reliability, reproducibility, and generalization performance. Multimodal EO datasets integrate heterogeneous sources-optical and multispectral imagery, Synthetic Aperture Radar (SAR), LiDAR-derived elevation models, in-situ sensor readings, vector GIS layers, and socio-economic records-capturing diverse spatial, temporal, and semantic relationships [4, 24]. They vary in geographic scale (local to global), spatial resolution (sub-meter to kilometer), and temporal frequency (daily to seasonal or annual) [3, 23]. Preprocessing pipelines typically involve spatial alignment, radiometric or atmospheric correction, noise filtering, normalization, feature extraction, and temporal synchronization. For EO data, augmentation strategies such as geometric transformations (rotation, scaling) and spectral band manipulation improve model robustness [18, 11]. Domain-specific priors-such as hydrological cycles in flood modelling or urban zoning semantics-are increasingly embedded into preprocessing to enhance interpretability and downstream performance [2, 10] (Table 6).

Each dataset poses unique preprocessing challenges. Optical–SAR combinations require precise registration and resolution harmonization to reconcile differences in acquisition geometry [4, 24]. Time-series datasets must address missing observations, irregular sampling, and seasonal variability [6, 23]. Integrating socio-economic or sensor data often demands sophisticated fusion techniques to align disparate spatial and temporal reference frames [18, 3]. To promote reproducibility, recent work emphasises standardized preprocessing protocols and open-source workflows [9, 11]. These toolkits

Table 5 Common GNN tasks in Earth Observation with modalities used

Task type	Typical modalities	References
Classification	Optical, SAR, LiDAR, OSM	[30, 10, 20]
Segmentation / Detection	Optical + SAR, DEM	[23, 20, 31]
Forecasting / Prediction	Optical time series, IoT, climate data	[2, 6, 8]
Anomaly / Vulnerability Detection	Urban imagery, IoT, socio-economic data	[9, 33, 28]
Change Detection	Multi-temporal optical + SAR, DEM	[23, 20, 31]
Resource & Environment Monitoring	Multispectral imagery, air quality sensors, socio-economic data	[1, 3, 2]

Table 6 Representative multimodal benchmark datasets with metadata and common preprocessing steps (illustrative)

Dataset	Modalities	Application domain	Year	License	Spatial extent	Typical preprocessing
Swiss Land Use Statistics [4]	Multispectral imagery, GIS land-use maps, census data	Land-use classification, change detection	2025	Open (CC BY 4.0)	Switzerland (nationwide, 1 ha units)	Georeferencing, normalization, label harmonization, noise filtering
M3LEO [24]	Interferometric SAR, multispectral optical, topographic maps	Multi-label EO classification	2024	Open (MIT)	Global (multi-sensor tiles)	Coregistration, speckle reduction, radiometric calibration, cloud masking
Terra [23]	Satellite imagery, climate sensor data, elevation models	Environmental monitoring, climate analysis	2024	Research-only license	Global (daily–monthly composites)	Temporal alignment, gap filling, feature normalization, spatial resampling
Urban Environmental Hazards [3]	Satellite imagery, air-quality sensors, socio-economic indicators	Disaster monitoring, urban sustainability	2024	Open (CC BY-NC)	Major global cities (multi-year)	Sensor calibration, spatial interpolation, noise removal, multimodal fusion
Crop Yield Prediction [6]	Optical imagery, meteorological time series, soil properties	Agriculture, yield forecasting	2024	Academic-use only	India (regional, seasonal cycles)	Temporal normalization, anomaly detection, spatial aggregation, feature engineering

Table 7 Quantitative comparison of representative multimodal GNN methods in EO applications. Reported metrics are taken directly from the cited studies

Domain / Task	Modalities integrated	Model type	Region / Dataset	Reported metrics
Crop yield prediction [6]	Multispectral imagery + weather time-series + parcel graphs	Temporal GNN + meta-transformer	India	$R^2 = 0.84$,RMSE ?12% vs baseline
Groundwater modeling [8]	SAR + DEM + in-situ gauges	Spatio-temporal GNN with cross-modal attention	UK	NSE +15% vs CNN
Urban land-use mapping [10]	Optical imagery + socio-economic + census data	Uncertainty-aware GNN (hybrid fusion)	China	Accuracy = 91.3%, F1 = 0.89
Deforestation mapping [28]	Optical + SAR + ecological vectors	Knowledge-guided GNN	SE Asia	IoU +8% vs CNN
Flood forecasting [2]	Radar + hydrological sensors + DEM	ST-GNN	Europe (multi-basin)	RMSE ?10% vs LSTM baseline

automate core steps such as geometric correction, normalization, and augmentation, reducing entry barriers for researchers and enabling fair method comparison [40, 5]. In addition to cataloguing datasets and preprocessing workflows, it is equally important to examine how multimodal GNNs perform quantitatively on EO tasks. Table 7 synthesizes results from representative studies, reporting evaluation metrics such as Accuracy, F1-score, Intersection-over-Union (IoU), Root Mean Squared Error (RMSE), and Nash–Sutcliffe Efficiency (NSE). These comparisons demonstrate that multimodal GNNs consistently outperform unimodal or non-graph baselines, particularly in applications requiring integration of spatial topology and temporal dynamics (Table 8).

The combination of a well-curated benchmark dataset, a domain-aware preprocessing pipeline, and transparent performance reporting forms the foundation for advancing multimodal GNNs toward robust, operational solutions in environmental, urban, agricultural, and resource management applications.

Table 8 Key challenges and emerging solution directions for multimodal GNNs in Earth Observation

Challenge	Emerging solution directions
Scalability to large graphs	Use hierarchical pooling and sampling to reduce graph size, distributed GNN training for parallelization, and domain-aware graph partitioning that preserves geospatial hierarchies (pixels–parcels–regions). Hardware acceleration with GPU/TPU clusters can further enable training on continental-scale datasets.
Temporal misalignment	Apply event-driven GNNs that process sensor readings as irregular events, or multi-rate temporal modules that explicitly model asynchronous sampling. Adaptive attention mechanisms can prioritize modalities depending on availability and recency.
Lack of interpretability	Develop interpretable GNN architectures that provide modality-wise attribution and graph-level explanations. Combine these with visual analytics tools to allow domain experts to trace predictions back to their spatial and data sources.
Noise and imbalance	Introduce uncertainty-aware objectives that propagate confidence estimates, design robust loss functions that address rare but critical classes, and apply EO-specific graph augmentation (e.g., reflectance perturbations within physical limits).
Cross-domain generalization	Use adversarial domain adaptation, cross-sensor normalization, and meta-learning. Few-shot adaptation and multi-source pretraining may improve robustness when transferring to new regions or sensor types.
Multimodal fusion difficulties	Incorporate adaptive cross-modal attention to balance modalities, modality-drop-out to simulate missing sensors, and uncertainty-aware fusion layers that explicitly track and propagate confidence from each modality.
Ethical concerns	Implement fairness auditing frameworks, privacy-preserving graph learning (e.g., federated or encrypted computation), and carbon-aware training protocols that quantify and reduce environmental costs.

5 Fusion strategies in GNN architectures

The integration of heterogeneous Earth Observation (EO) data within Graph Neural Network (GNN) architectures has emerged as a central research theme due to the complementary nature of EO modalities. Fusion strategies are commonly categorized into three main types-*early fusion*, *late fusion*, and *hybrid fusion*-each offering distinct trade-offs in terms of complexity, robustness, interpretability, and capacity to capture cross-modal interactions [4, 23]. More recent approaches extend these paradigms through cross-modal attention, heterogeneous graph modeling, and the incorporation of pre-trained foundation models.

5.1 Early fusion

Early fusion concatenates raw or preprocessed features from multiple modalities prior to graph construction and message passing, enabling the network to jointly learn cross-modal feature interactions from the outset. This approach can yield rich embeddings and improved predictive accuracy when modalities are well aligned [4, 12]. For instance, in deforestation mapping, combining multispectral and SAR features at the pixel level has been shown to improve Intersection-over-Union (IoU) scores by 5–7% compared to unimodal baselines [28]. However, early fusion requires precise spatial–temporal registration, can be computationally expensive due to high input dimensionality, and is prone to feature dominance, where high-signal modalities (e.g., multispectral imagery) suppress weaker but complementary sources (e.g., sparse IoT data) [18].

5.2 Late fusion

Late fusion processes each modality in a separate GNN branch or subnetwork, combining outputs at a decision or embedding level. This modular design accommodates modality-specific preprocessing and noise characteristics, making it effective

for heterogeneous sources such as dynamic weather time-series and static land-cover maps [2, 23]. It improves robustness to missing or degraded modalities and supports interpretability by isolating modality contributions. For example, Yan et al. [10] applied late fusion to urban land-use classification and achieved an accuracy of 85.2%, but performance was limited compared to hybrid fusion strategies (see below). Its main drawback is reduced exploitation of fine-grained inter-modality interactions during early representation learning [9].

5.3 Hybrid fusion

Hybrid fusion integrates the strengths of both early and late fusion by performing modality-specific feature extraction before joint representation learning in later GNN layers. This design supports flexible encoding while enabling cross-modal interaction modeling in intermediate or deep stages [4, 5]. Case studies demonstrate that hybrid approaches consistently outperform simpler strategies: Sarkar et al. [6] reported that a hybrid temporal GNN for crop yield prediction reduced RMSE by 12% compared to late fusion baselines, while Taccari et al. [8] achieved a 15% gain in Nash–Sutcliffe Efficiency for groundwater modeling when replacing early fusion with attention-based hybrid fusion. Similarly, Yan et al. [10] observed F1-score improvements of 6–8% in urban mapping tasks using uncertainty-aware hybrid models. These results highlight that hybrid and attention-based multimodal GNNs deliver superior accuracy and robustness, albeit with increased architectural complexity and higher risk of overfitting [22].

5.4 Trade-offs and design considerations

Fusion strategy selection depends on factors such as dataset scale, modality heterogeneity, alignment quality, and computational constraints. Early fusion is best when strong cross-modal correlations and precise registration are available; late fusion excels with asynchronous, noisy, or incomplete modalities; hybrid fusion offers the most balanced flexibility but demands careful architectural tuning. Persistent challenges include mismatched resolutions, missing data, and sensor-specific noise, motivating research into adaptive fusion mechanisms, dynamic modality weighting, and architectures resilient to real-world EO data imperfections [3, 2]. Beyond the three classical paradigms, recent work leverages cross-modal attention, heterogeneous graphs, and pretrained vision–language–geospatial models to improve generalization and interpretability across diverse EO tasks [22, 7].

These trade-offs are illustrated in Fig. 2, which schematically compares the three main fusion strategies in multimodal GNN architectures.

6 Challenges and open research problems

The integration of multimodal Earth Observation (EO) data with Graph Neural Networks (GNNs) offers enormous potential, but it also faces several unresolved challenges that limit scalability, generalizability, and responsible deployment. Unlike unimodal or grid-based EO models, multimodal GNNs must simultaneously process heterogeneous sensors, asynchronous acquisitions, irregular spatial footprints, and distribution shifts across regions and sensors. These difficulties give rise to a range of technical, methodological, and ethical research problems that must be addressed before multimodal

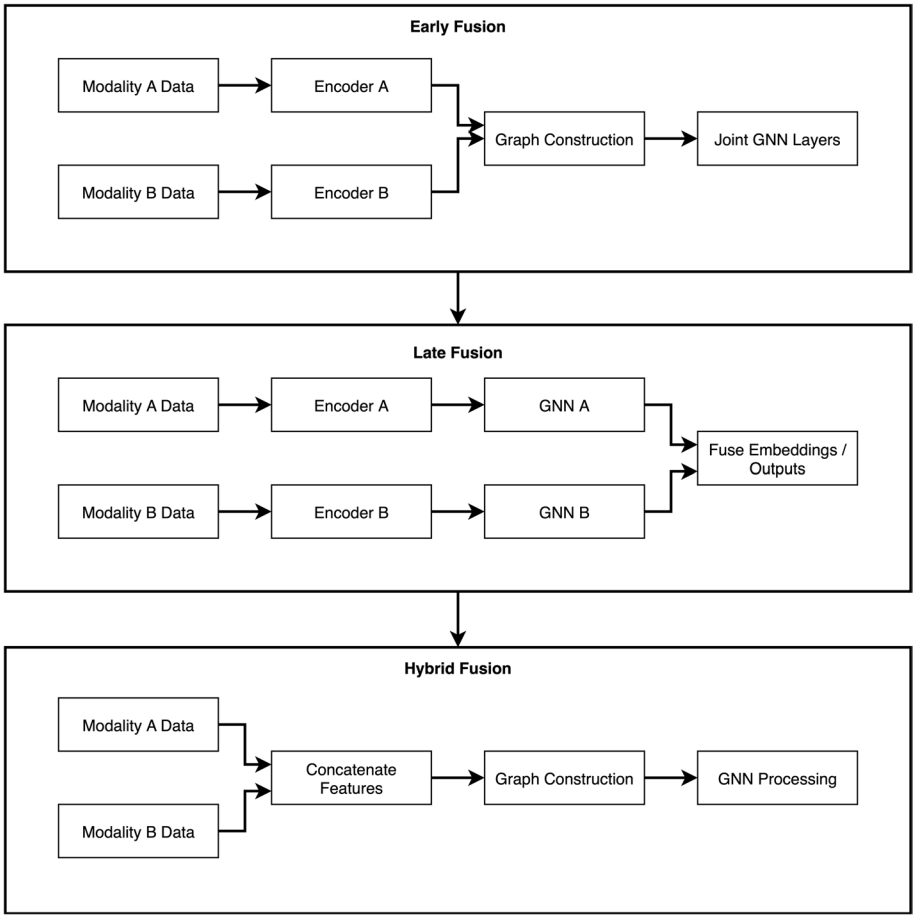

Fig. 2 Comparison of fusion strategies for multimodal GNN architectures. Early fusion integrates modalities before graph construction, late fusion combines outputs from separate branches, and hybrid fusion allows modality-specific processing with subsequent joint learning

GNNs can become operational tools for sustainability, disaster management, and policy support [4, 12, 23]. Below, we organize these challenges into key categories.

6.1 Scalability to large and complex graphs

EO systems generate vast data streams from multispectral imagery, SAR, LiDAR, IoT sensors, and socio-economic sources. Representing these data as graphs often leads to millions of nodes and edges, pushing memory and computational limits [39, 5]. The problem intensifies in multimodal settings, where each modality introduces additional node and edge types, increasing graph density. Current scalability solutions such as hierarchical pooling, sampling, and distributed GNN training [18] have mainly been developed for social networks and citation graphs, where topological constraints are weaker. EO-specific challenges remain unresolved: how to preserve multi-resolution spatial hierarchies (pixels–parcels–regions), how to respect physical adjacency (watersheds, road networks), and how to maintain geodetic accuracy at scale. Domain-aware graph partitioning strategies, hardware acceleration (GPU/TPU clusters), and hybrid cloud–edge infrastructures will be required to scale models to continental or global EO datasets.

6.2 Temporal alignment and irregular sampling

Different EO modalities operate at inherently different temporal resolutions: optical satellites revisit in 5–10 days, SAR in 6–12 days, and in-situ IoT sensors stream data hourly or irregularly [4, 6]. Cloud cover, maintenance, or outages further disrupt continuity. Aligning these asynchronous time series is critical, yet existing temporal GNNs and transformer-based encoders [5] usually assume regularly sampled data and rely on interpolation or resampling, which risks distorting dynamics. The difficulty increases when fusing long-term climate trends with short-lived extreme events (floods, landslides). Promising directions include event-driven GNNs that model sensor readings as discrete events, multi-rate temporal modules that handle different sampling frequencies in parallel, and adaptive attention mechanisms that dynamically weight available modalities by quality and recency.

6.3 Explainability and interpretability

When EO-based predictions are used for disaster response, resource allocation, or policy planning, trust and accountability are essential [2]. Yet multimodal GNNs, which combine spatio-temporal graphs across heterogeneous sensors, are particularly difficult to interpret. Existing explanation methods (e.g., attention visualization, gradient-based attribution) struggle to disentangle modality-specific and spatial contributions [4]. For example, a flood risk model combining SAR backscatter and census data may be accurate, but decision-makers need to know whether predictions are driven by topography, infrastructure density, or socio-economic features. Progress is needed in interpretable architectures that provide modality-wise attribution, graph-level explanations, and visual analytics tools tailored to EO. Without such tools, real-world adoption will remain slow.

6.4 Noise, missing data, and class imbalance

EO datasets are noisy and incomplete: optical imagery suffers from clouds, SAR from speckle, and in-situ sensors from drift or outages. Rare but critical events (wildfires, landslides) produce severe class imbalance [3]. Conventional preprocessing-denoising, imputation, oversampling-may erase important variability or introduce bias [23]. Missing modalities further complicate graph construction: when entire sensors or layers are absent, relational modeling is degraded. Promising strategies include uncertainty-aware training objectives that propagate confidence estimates through the graph, EO-specific graph augmentation (e.g., perturbing reflectance values within physical limits), and loss functions designed to rebalance rare but high-impact classes.

6.5 Domain adaptation and generalization

GNNs trained in one region or sensor setup often fail in others due to distribution shifts in climate, land cover, or acquisition parameters. For example, a crop classification model trained on Sentinel-2 data in Europe may degrade when applied to Asian croplands, or when switching from Sentinel to commercial satellites [4]. Existing domain adaptation strategies-adversarial training, meta-learning, cross-domain pretraining-are underexplored in multimodal EO pipelines [23]. Recent works highlight potential, but general-purpose, transferable multimodal GNNs remain elusive. Achieving robust cross-sensor and cross-region generalization is a prerequisite for global EO analytics.

6.6 Challenges in multimodal fusion

Constructing graphs that jointly represent pixels, parcels, sensors, and socio-economic units without collapsing semantics remains difficult. High-resolution imagery often dominates fusion, overshadowing weaker but semantically rich modalities (e.g., air quality readings, census statistics). Missing modalities further unbalance the graph. Errors in one modality may also cascade across edges, leading to unstable predictions. Emerging solutions include adaptive cross-modal attention (dynamically balancing sources), modality-dropout (simulating missing inputs during training), and uncertainty-aware fusion that propagates per-modality confidence.

6.7 Ethical and responsible AI

Beyond technical issues, ethical challenges must be treated as central to research design [29]. Training large multimodal GNNs requires massive compute, raising environmental concerns about carbon emissions. Quantitative metrics such as energy-per-training-step or model carbon footprint are rarely reported but should become standard. Privacy risks arise when satellite data are linked with census or mobility datasets, potentially exposing sensitive populations [2]. Biases in EO data-geographic, demographic, or sensor-related-may propagate to downstream predictions, affecting vulnerable communities. Responsible multimodal EO requires fairness auditing, privacy-preserving computation (federated learning, secure multiparty computation), and carbon-aware training protocols. Without explicit ethical safeguards, operational deployment risks doing more harm than good.

The interdependencies among these challenges are illustrated in Fig. 3, highlighting how scalability, fusion, interpretability, and ethics are tightly interconnected.

Multimodal GNNs in EO face intertwined challenges in scalability, temporal fusion, interpretability, data quality, generalization, and responsible AI. These limitations form the foundation for the research directions discussed in the next section.

7 Future directions

Building on the limitations identified in Sect. 6, this section highlights emerging research directions to overcome current obstacles and unlock the full potential of multimodal GNNs in Earth Observation (EO). The key priorities include self-supervised learning, dynamic graph modeling, edge intelligence, explainability, fusion strategies, and cross-domain adaptation. Together, these constitute a roadmap toward operational, trustworthy, and sustainable EO applications [4, 39].

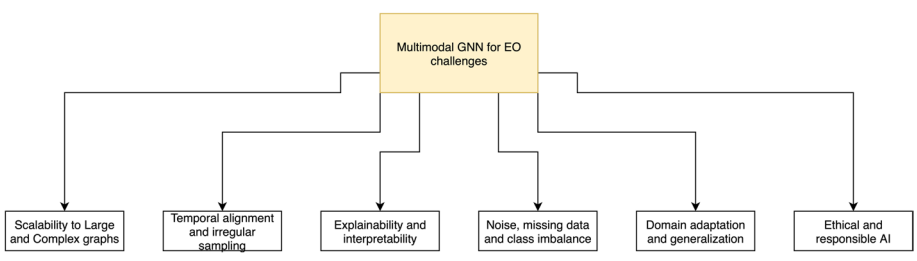

Fig. 3 Interrelated challenges and open research problems in multimodal GNNs for EO

7.1 Self-supervised and foundation models for multimodal EO

Self-supervised learning (SSL) and foundation models are rapidly transforming EO analytics [22, 41]. SSL leverages large unlabeled corpora using proxy tasks such as contrastive learning, masked prediction, or reconstruction, thereby reducing dependence on costly annotations. Foundation models pre-trained on multimodal EO datasets can be fine-tuned for downstream tasks, enabling rapid adaptation to diverse sensors and geographies. When coupled with GNNs, such approaches simultaneously capture spatial, temporal, and semantic dependencies. Early results in agriculture report reductions of 10–12% in RMSE for yield prediction tasks compared to conventional baselines [6], underscoring their potential for crop monitoring, land-use mapping, and climate-risk assessment.

7.2 Integration of dynamic and evolving graph structures

EO systems are inherently dynamic, as urban expansion, deforestation, and climate change continually reshape spatial relations. Static graphs cannot capture these evolving dependencies. Dynamic GNN architectures, where nodes, edges, and topology evolve over time, are needed to model real-world environmental processes [5]. Spatio-temporal GNNs and temporal graph attention networks are early steps, but scalability and uncertainty handling remain open problems. Incorporating dynamic structures has already improved hydrological forecasting by up to 15% in Nash–Sutcliffe Efficiency over CNNs [8], suggesting strong potential for biodiversity monitoring, disaster prediction, and infrastructure resilience.

7.3 Edge computing and real-time EO analytics

Real-time EO analytics are critical for disaster response, traffic monitoring, and precision agriculture [42, 25]. Centralized cloud pipelines often introduce latency due to data transfer bottlenecks. Edge computing provides an alternative by running lightweight GNN inference directly on satellites, UAVs, or IoT sensors. Techniques such as pruning, quantization, and distributed training can deliver low-latency predictions under resource constraints [39]. Hybrid cloud–edge ecosystems further enable scalable yet responsive EO decision support, with latency reductions of up to 40% reported in prototype systems.

7.4 Enhancing explainability and trustworthiness

Trustworthy and interpretable AI is essential for EO applications with policy or disaster-management implications. Current black-box multimodal GNNs limit adoption by domain experts. Future research should establish benchmarks for interpretability and develop models that provide modality-specific attribution, causal inference, and visual analytics [2]. For example, uncertainty-aware GNNs in urban land-use classification achieved an F1-score of 0.89 while providing per-modality attribution [10]. Embedding such interpretability within multimodal GNN pipelines will increase stakeholder trust and facilitate responsible deployment.

7.5 Integration of EO, IoT, and other data sources with GNNs

The rapid expansion of Internet-of-Things (IoT) sensor networks, crowd-sourced geospatial data, and socio-economic information has created new opportunities for

multimodal Graph Neural Networks (GNNs) in Earth Observation (EO). Integrating these complementary sources with satellite and aerial imagery can provide richer, context-aware representations of environmental and human systems, though it also introduces several technical challenges. A common strategy is to construct heterogeneous graphs where nodes represent EO pixels or parcels, IoT sensors, and socio-economic entities, while edges encode spatial proximity, functional similarity, or dynamic interactions such as resource flow. Cross-modal attention mechanisms and modality-specific encoders enable effective fusion by adaptively weighting asynchronous or noisy modalities. Hybrid edge–cloud architectures are also emerging, in which IoT devices supply real-time sensor data for lightweight GNN inference at the edge, while computationally intensive satellite-image processing occurs in the cloud. Integrating EO and IoT data improves spatio-temporal resolution and robustness. For example, soil moisture and air quality sensors can complement multispectral imagery for precision agriculture and urban environmental monitoring. Similarly, socio-economic and mobility datasets enrich contextual understanding, supporting more reliable assessments of vulnerability, infrastructure resilience, and policy effectiveness. Despite these advantages, integration remains challenging due to heterogeneous temporal sampling-IoT sensors typically stream at high frequency while satellite systems operate at longer revisit intervals. Additional concerns include privacy when linking geospatial and demographic data, interoperability across EO and IoT standards, and scalability issues when millions of sensors must be represented as nodes within a graph. Addressing these constraints will be essential for realizing the full potential of GNN-based fusion across EO and IoT ecosystems.

7.6 Advances in multimodal fusion and cross-domain adaptation

Fusion remains the core challenge in multimodal EO. Adaptive mechanisms that dynamically weigh modalities by quality and availability can improve robustness to noise or missing data [4, 23]. Hybrid and attention-based fusion has already demonstrated 6–8% higher F1-scores compared to late fusion in urban mapping [10]. At the same time, cross-domain adaptation is critical for global EO applications. Techniques such as adversarial learning, meta-learning, and few-shot transfer have shown promise in handling domain shifts across regions and sensors, but general-purpose multimodal GNNs remain elusive. These emerging directions are summarized in Table 9, which maps each research priority to enabling technologies and the corresponding expected benefits. The

Table 9 Mapping of future research directions to enabling technologies and expected benefits

Future direction	Enabling technologies	Expected benefits
Self-supervised and foundation models	Contrastive learning, masked prediction, multimodal pretraining	Reduced labeling costs, improved generalization, rapid adaptation to new sensors and regions
Dynamic graph modeling	Spatio-temporal GNNs, temporal attention, graph evolution modules	Accurate change detection, improved temporal resolution, real-time monitoring of evolving phenomena
Edge computing for real-time EO	Model compression, pruning, distributed inference on satellites/UAVs	Low-latency analytics, bandwidth efficiency, rapid response in time-critical scenarios
Explainability and trustworthiness	Attention visualization, causal inference, interpretability benchmarks	Improved transparency, higher stakeholder trust, responsible policy adoption
Advanced multimodal fusion	Adaptive weighting, cross-modal transformers, uncertainty modeling	Robustness to noisy/missing modalities, more accurate multimodal integration
Cross-domain adaptation	Adversarial training, transfer learning, few-shot domain adaptation	Enhanced generalization across regions and sensors, reduced vulnerability to domain shifts

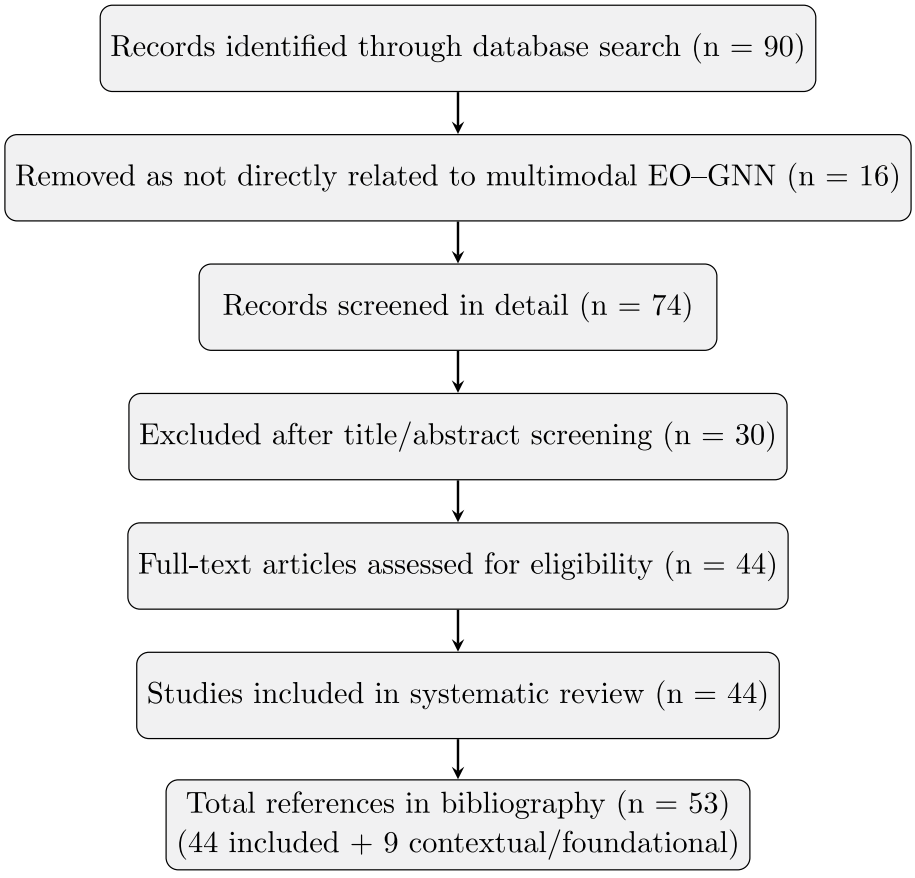

Fig. 4 PRISMA-style flow diagram of the paper selection process. Out of 90 records retrieved, 18 were excluded as not directly related to multimodal EO–GNN. After detailed screening and eligibility checks, 42 studies were included. An additional 9 contextual references were cited, yielding a total of 51 references

table provides a consolidated view of how future methods can bridge current limitations and advance multimodal EO-GNN applications (Fig. 4).

7.7 Key takeaways

- SSL and foundation models reduce reliance on scarce labeled EO datasets.
- Dynamic and evolving graph structures improve modeling of rapidly changing environments.
- Edge intelligence enables low-latency decision support in disaster and precision-agriculture contexts.
- Explainability and responsible AI are prerequisites for adoption in policy and operational settings.
- Adaptive fusion and domain adaptation are essential for robust, global-scale EO-GNN applications.

The next phase of research in multimodal EO-GNNs will thus be driven by advances in foundation learning, dynamic graph models, edge intelligence, interpretability, and adaptive fusion. Collectively, these priorities aim to bridge current limitations, accelerate real-world adoption, and ensure that EO analytics contribute effectively to sustainability and resilience goals.

8 Conclusion

This survey has provided a comprehensive review of multimodal Graph Neural Networks (GNNs) for Earth Observation (EO), with a particular focus on their role in advancing sustainability-related monitoring and decision-making. We introduced a structured taxonomy of fusion strategies-including early, late, hybrid, cross-modal attention, and emerging foundation-model-based approaches-and analyzed representative architectures deployed across agriculture, water resources, forestry, urban infrastructure, and related domains. Public datasets, benchmark tasks, and evaluation metrics were systematically compared, and critical challenges such as label scarcity, domain shift, scalability, interpretability, uncertainty quantification, and real-time deployment were examined in detail. Through this synthesis, the survey highlights both the methodological maturity and the practical limitations of current multimodal GNNs for EO. Despite recent progress, several gaps remain. Scalable architectures that can efficiently handle multi-resolution, multi-sensor graphs are still lacking. Adaptive and robust fusion mechanisms capable of coping with noisy, asynchronous, and missing modalities remain underdeveloped. Furthermore, integrating physics-informed priors and uncertainty-aware modeling into graph-based pipelines is still in its infancy, while the environmental footprint of large models underscores the need for energy-efficient "green AI" practices. These issues must be addressed to ensure the responsible and operational deployment of multimodal GNNs at global scale. Looking ahead, we identify five priority directions for future research:

(i) Developing self-supervised and transfer learning methods to reduce reliance on scarce labeled datasets;
(ii) Designing dynamic and streaming graph models for continuous EO monitoring;
(iii) Integrating interpretable and uncertainty-aware AI components to enhance trust in decision support;
(iv) Leveraging edge and hybrid cloud–edge infrastructures for scalable, low-latency analytics; and
(v) Establishing standardized benchmarks, shared datasets, and reproducible protocols to accelerate fair and transparent evaluation.

By consolidating methodological advances, domain-specific applications, benchmark resources, and unresolved challenges, this survey provides a roadmap for the next generation of multimodal GNN systems. With coordinated efforts across EO science, machine learning, and sustainability research, multimodal GNNs hold the potential to deliver actionable, policy-relevant insights that contribute directly to global priorities in climate resilience, disaster preparedness, food security, and sustainable urbanization.

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Author contributions
H.S. conducted the literature review and drafted the main manuscript text. S.K. prepared Figs. 1, 2 and 3, organized the taxonomy, and contributed to writing and editing. Both authors reviewed and approved the final manuscript.

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Code availability
Not applicable.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent to publication
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Human or animal rights
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References

1. Nie J, Zuo Z, Liang X, Ye M, Song N, Li X. Multimodal issues and key technologies in ocean observation data: a review. IEEE J Sel Top Appl Earth Obs Remote Sens. 2025;8:45.
2. Biazar SM, Golmohammadi G, Nedhunuri RR, Shaghaghi S, Mohammadi K. Artificial intelligence in hydrology: advancements in soil, water resource management, and sustainable development. Sustainability. 2025;17(5):2250.
3. Kumar D, Bassill NP, Ghosh S. Analyzing recent trends in deep-learning approaches: a review on urban environmental hazards and disaster studies for monitoring, management, and mitigation toward sustainability. 2024;
4. Meyer A, Milani G, Lack N, Schönholzer M, Jordan D. Machine learning-based multimodal data fusion for the Swiss land use statistics. Eur J Remote Sens. 2025;58(1):2508420.
5. Dhivya K, Kumar SN, Victoria DRS, Sherly SI, Durgadevi G. Advanced neural networks for multimodal data fusion in interdisciplinary research. In: Advanced Interdisciplinary Applications of Deep Learning for Data Science, 2025 pp. 201–232. IGI Global Scientific Publishing
6. Sarkar S, Dey A, Pradhan R, Sarkar UM, Chatterjee C, Mondal A, Mitra P. Crop yield prediction using multimodal meta-transformer and temporal graph neural networks. IEEE Transactions on Agrifood Electronics 2024;
7. Yang ZX, Li Y, Wang RF, Hu P. Deep learning in multimodal fusion for sustainable plant care: a comprehensive review. Sustainability. 2025;17:12.
8. Taccari ML, Wang H, Nuttall J, Chen X, Jimack PK. Spatial-temporal graph neural networks for groundwater data. Sci Rep. 2024;14(1):24564.
9. Amangeldy B, Imankulov T, Tasmurzayev N, Dikhanbayeva G, Nurakhov Y. A review of artificial intelligence and deep learning approaches for resource management in smart buildings. Buildings. 2025;15(15):2631.
10. Yan X, Jiang Z, Luo P, Wu H, Dong A, Mao F, et al. A multimodal data fusion model for accurate and interpretable urban land use mapping with uncertainty analysis. Int J Appl Earth Obs Geoinf. 2024;129:103805.
11. Sheikh N, Saha S. Graph neural networks for multi-sensor Earth observation. In: Deep Learning for Multi-Sensor Earth Observation, 2025 pp. 211–230
12. Zhao S, Chen Z, Xiong Z, Shi Y, Saha S, Zhu XX. Beyond grid data: exploring graph neural networks for Earth observation. IEEE Geoscience and Remote Sensing Magazine 2024;
13. Benedetti P, Ienco D, Gaetano R, Ose K, Pensa RG, Dupuy S. M^3Fusion: a deep learning architecture for multiscale multimodal multitemporal satellite data fusion. IEEE J Sel Top Appl Earth Obs Remote Sens. 2018;11(12):4939–49.
14. Persello C, Wegner JD, Hänsch R, Tuia D, Ghamisi P, Koeva M, et al. Deep learning and earth observation to support the sustainable development goals: current approaches, open challenges, and future opportunities. IEEE Geosci Remote Sens Mag. 2022;10(2):172–200.
15. Lemmens R, Lang S, Albrecht F, Augustijn E, Granell C, Olijslagers M, Barton J. Integrating concepts of artificial intelligence in the EO4GEO body of knowledge. In: *XXIV ISPRS Congress Imaging Today, Foreseeing Tomorrow, Commission IV*, 2022;**43**, pp. 53–59. Copernicus Publications.
16. Wang Z, Li Z, Wang B, Liu H. Robot grasp detection using multimodal deep convolutional neural networks. Adv Mech Eng. 2016;8(9):1687814016668077.
17. Hagen E, Næss S, Ness TV, Einevoll GT. Multimodal modeling of neural network activity: computing LFP, ECoG, EEG, and MEG signals with LFPy 2.0. Front Neuroinform. 2018;12:92.
18. Bhat VP, Harsha TR, Vinay SK. Revolutionizing AI: Applications of graph neural networks across industries. In: Graph Neural Networks: Essentials and Use Cases, 2025 pp. 161–188. Springer Nature, Cham
19. Liu Z, Zhang J, Shi E, Liu Z, Niyato D, Ai B, Shen XS. Graph neural network meets multi-agent reinforcement learning: Fundamentals, applications, and future directions. *IEEE Wireless Communications* 2024;
20. Wang M, Li X, Tan K, Mango J, Pan C, Zhang D. Position-aware Graph-CNN fusion network: an integrated approach combining geospatial information and graph attention network for multiclass change detection. IEEE Trans Geosci Remote Sens. 2024;62:1–16.
21. Yang Q, Giezendanner J, Civitarese DS, Jakubik J, Schmitt E, Chandra A, Wang S. Multi-modal graph neural networks for localized off-grid weather forecasting 2024;
22. Chen Y, Hu B, Liu Y. Optimizing document management and retrieval with multimodal transformers and knowledge graphs. PLoS One. 2025;20(6):e0323966.

23. Chen B, Gao Z, Li Z, Liu S, Hu A, Song W, et al. Hierarchical GNN framework for earth's surface anomaly detection in single satellite imagery. IEEE Trans Geosci Remote Sens. 2024;62:1–14.
24. Allen M, Dorr F, Gallego Mejia JA, Martínez-Ferrer L, Jungbluth A, Kalaitzis F, et al. M3leo: a multi-modal, multi-label earth observation dataset integrating interferometric SAR and multispectral data. Adv Neural Inf Process Syst. 2024;37:104694–723.
25. Dhanasekaran S, Gopal D, Logeshwaran J, Ramya N, Salau AO. Multi-model traffic forecasting in smart cities using graph neural networks and transformer-based multi-source visual fusion for intelligent transportation management. Int J Intell Transp Syst Res. 2024;22(3):518–41.
26. Dong G, Tang M, Wang Z, Gao J, Guo S, Cai L, et al. Graph neural networks in IoT: a survey. ACM Trans Sensor Netw. 2023. https://doi.org/10.1145/3565973.
27. Chatrabhuj Meshram K. Incremental learning model for sustainable agricultural land assessment using multimodal satellite data. Int J Remote Sens. 2024;45:8622–48. https://doi.org/10.1080/01431161.2024.2403628.
28. Liu H, Ding Q, Yang X, Liu Q, Deng M, Gui R. A knowledge-guided approach for landslide susceptibility mapping using convolutional neural network and graph contrastive learning. Sustainability. 2024;16(11):4547.
29. Ghamisi P, Yu W, Marinoni A, Gevaert CM, Persello C, Selvakumaran S, Atkinson PM. Responsible Artificial Intelligence for Earth Observation: Achievable and realistic paths to serve the collective good. *IEEE Geoscience and Remote Sensing Magazine* 2025;
30. Lou K. Li M, Li F, Zheng X. Integrating local–global structural interaction using Siamese graph neural network for urban land use change detection from VHR satellite images. *IEEE Transactions on Geoscience and Remote Sensing* 2024;
31. Khan MA, Hamza A, Ibrar W, Jamel L, Alasiry A, Marzougui M, Nam Y. Coastal and Land Use Land Cover Area Recognition from High-Resolution Remote Sensing Images using a Novel Multimodal Attention Inception Residual Deep Network. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 2025;
32. MirhoseiniNejad SM, Abbasi-Moghadam D, Sharifi A. ConvLSTM-ViT: a deep neural network for crop yield prediction using Earth observations and remotely sensed data. IEEE J Sel Top Appl Earth Obs Remote Sens. 2024;20:24.
33. Thangamayan S, Ramu M, Sundaram A. Deep learning-enhanced remote sensing for analysing socio-economic impacts of climate change: an interdisciplinary perspective. Remote Sens Earth Syst Sci. 2024;7(4):612–26.
34. Zhang P, Yang M, Wang Y, Yang T, Yu H, Yan X. Integrating metro passenger flow data to improve the classification of urban functional regions using a heterogeneous graph neural network. Int J Digit Earth. 2024;17(1):2443468.
35. Jiang W, Zhang Y, Han H, Huang Z, Li Q, Mu J. Mobile traffic prediction in consumer applications: a multimodal deep learning approach. IEEE Trans Consum Electron. 2024;70(1):3425–35.
36. Zheng J, Li G, Wen Y, Zhang J, Dong R, Fu H. "Evidential Graph Contrastive Alignment for Source-Free Blending-Target Domain Adaptation," *IEEE Transactions on Neural Networks and Learning Systems*, 2025;
37. Chappidi J, Sundaram DM. Dual Q-Learning with graph neural networks: a novel approach to animal detection in challenging ecosystems. *Journal of Theoretical and Applied Information Technology*2024; **102**(23)
38. Li Q, Zhang Y, Zheng J, Zhang Y, Huang J, Fu H. "Boosting Universal Domain Adaptation in Remote Sensing With Dual-Classifiers Consistency Discrimination and Cross Domain Feature Mixup,"*IEEE Transactions on Geoscience and Remote Sensing*, 2025;(63)1-15, 5515015 https://doi.org/10.1109/TGRS.2025.3571747.
39. Sun R, Cheng N, Li C, Quan W, Zhou H, Wang Y, et al. A comprehensive survey of knowledge-driven deep learning for intelligent wireless network optimization in 6G. IEEE Communications Surveys & Tutorials 2025;
40. Garcia-Palencia O, Fernandez J, Shim V, Kasabov NK, Wang A. Alzheimer's Disease Neuroimaging Initiative: Spiking neural networks for multimodal neuroimaging: a comprehensive review of current trends and the NeuCube brain-inspired architecture. Bioengineering. 2025;12(6):628.
41. Zhang W, Bei Y, Yang L, Zou HP, Zhou P, Liu A, et al. Cold-start recommendation towards the era of large language models (llms): a comprehensive survey and roadmap. arXiv preprint arXiv:2501.01945 2025;
42. Ashqar HI, Jaber A, Alhadidi TI, Elhenawy M. Advancing object detection in transportation with multimodal large language models (mllms): a comprehensive review and empirical testing. Computation. 2025;13(6):133.
45. Chatrabhuj MK. Proposed design of an augmented deep learning model for estimation of sustainable development goals via satellite image analysis. Environ Dev Sustain. 2025;27:11307–33.
46. Chatrabhuj Meshram K. Design of an efficient satellite image analysis model for identification of sustainable construction areas via ground subsidence monitoring and infrastructure monitoring operations. Indian Geotech J. 2024;54:1136–51. https://doi.org/10.1007/s40098-024-00907-8.
50. Park SM, Kim YG. A metaverse: taxonomy, components, applications, and open challenges. IEEE Access. 2022;10:4209–51.

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